

MUHAMMAD ALI

SENIOR SOFTWARE ENGINEER | Node | TypeScript | AWS | Azure

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Summary

Experienced Node.js Developer with 8+ years of expertise in designing and building scalable, high-performance backend systems using Node.js, Express.js, React, and NestJS, with a proven track record in architecting microservices and event-driven systems, and developing robust REST, GraphQL, and gRPC APIs. Strong experience in real-time applications using WebSockets/Socket.IO and Redis, along with hands-on expertise in cloud platforms (AWS, GCP, Azure), Docker, Kubernetes, and CI/CD pipelines. Passionate about writing clean, maintainable, and efficient code using modern JavaScript and TypeScript, with a strong focus on performance, scalability, and system reliability.

Skills

Frontend: React.js, Next.js, Angular, Tailwind CSS, TypeScript, State Management (Redux/Zustand).

Backend: Node.js (NestJS, Express.js), Python (FastAPI, Django), RESTful APIs, GraphQL, Microservices Architecture, WebSockets (Socket.io)

Database: PostgreSQL, MongoDB, SQL Server, MySQL, DynamoDB, Query Optimization, Schema Design.

Infra & Cloud: AWS (Lambda, S3, ECS, API Gateway, Aurora), Azure, Google Cloud Platform (GCP), Docker, Kubernetes, Serverless Architecture, Redis Caching.

Testing: Unit & Integration Testing, Jest, Mocha, Chai, React Testing Library, Pytest, Cypress, TDD (Test Driven Development)

CI/CD & DevOps: GitHub Actions, Jenkins, GitLab CI, CI/CD Pipeline Automation, Infrastructure as Code (Terraform), Version Control (Git).

Experience

Senior Software Engineer

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12/2021 - Present

- A full-service Agile Software Development company focused on delivering innovative solutions.
- Designed and architected scalable RESTful APIs using Node.js and NestJS, ensuring secure and high-performance communication between microservices and external integrations.
- Spearheaded the migration of legacy monolithic services to a Microservices architecture, improving system modularity and reducing application runtime by 40%
- Developed and maintained full-stack web applications using JavaScript/TypeScript, focusing on Node.js for backend and Next.js for server-side rendering.
- Implemented CI/CD pipelines using GitHub Actions and AWS CodePipeline, automating testing and deployment processes, which increased deployment frequency by 25%.
- Wrote clean, efficient, and well-documented code that adheres to industry best practices and standards.
- Utilized TypeORM and Mongoose to design complex database schemas in PostgreSQL and MongoDB, optimizing query execution and data access layers for high-speed retrieval.
- Built highly interactive and reactive frontend dashboards using React/Next.js (with Redux and RxJS), significantly improving user data interactivity and frontend performance.
- Implemented secure authentication and authorization flows using Passport.js, JWT (JSON Web Tokens), and Role-Based Access Control (RBAC).
- Integrated AWS S3 (Simple Storage Service) for secure, large-scale file and imagery uploads, implementing Presigned URLs for secure data access.
- Architected modular API endpoints using a Layered Architecture (Controller, Service, Repository patterns) to ensure code maintainability and clean separation of concerns.
- Designed complex SQL queries, indexes, and stored procedures in SQL Server and managed NoSQL document storage in MongoDB.
- Wrote comprehensive unit and integration tests using Jest and Supertest, achieving 95% code coverage and ensuring production environment stability.
- Integrated Azure Blob Storage for secure, large-scale drone imagery and file uploads with presigned URL handling.
- Contributed to the migration of the database from SQL Server to Azure SQL Database, streamlining data storage, security, and access for remote teams.

Experience

Senior Software Engineer

Pano AI

07/2020 - 11/2021

- Specializes in building solutions for the hospitality industry using advanced technology.
- Spearheaded the development of a real-time recommendation engine using **collaborative filtering** and **Node.js**, improving user engagement by **30%**.
- Designed and developed scalable backend solutions using **Node.js and Express.js**, ensuring optimal performance and low latency for high-traffic applications.
- Architected and implemented RESTful APIs and GraphQL APIs, ensuring smooth data exchange between the frontend and backend.
- Integrated third-party services such as payment gateways, email services, and file storage solutions to extend the functionality of applications.
- Developed microservices using Docker and Kubernetes to enhance the modularity and scalability of the application infrastructure.
- Led efforts in building secure authentication systems with JWT and OAuth2.0 protocols, improving application security and user experience.
- Optimized database performance by designing efficient MongoDB and PostgreSQL schemas, reducing query response times by 30%.
- Collaborated with front-end developers to integrate Node.js backends with modern front-end frameworks like React.js and Vue.js.
- Wrote unit and integration tests using Mocha and Chai, ensuring high code coverage and reliability.

Software Engineer

Devsinc

01/2017 - 06/2020

- Develops solutions for secure securities filings
- Assisted senior developers in building and maintaining **scalable Node.js and Express.js** applications, focusing on high-performance backend architecture.
- Developed server-side logic and RESTful APIs to support multiple internal tools and external customer-facing applications.
- Maintained and optimized **JavaScript/TypeScript** codebases, refactoring legacy modules to improve maintainability and system performance.
- Participated in the development of real-time applications utilizing **WebSockets (Socket.io)** for seamless bi-directional communication.
- Automated complex data processing tasks using **Node.js scripts**, reducing manual intervention and increasing productivity by **40%**.
- Contributed to debugging, troubleshooting, and resolving high-priority issues in both development and production environments.
- Designed efficient **NoSQL schemas in MongoDB** and optimized **PostgreSQL** queries and indexes for high-speed data retrieval.
- Implemented secure authentication and authorization flows using **Passport.js, JWT**, and Role-Based Access Control (**RBAC**).

Education

California State University-Dominguez Hills

06/2010 - 07/2014

BS - Software Engineering